

# ECON 470 Homework 3

Submission 3, Spring 2026

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## Problem 1

Table 1: Summary Statistics by Year

| Year | Mean Star Rating | Mean Enrollments | Mean Market Share | Total Plans |
|------|------------------|------------------|-------------------|-------------|
| 2010 | 2.97             | 256.35           | 0.0658            | 108222      |
| 2011 | 3.26             | 343.36           | 0.0855            | 68003       |
| 2012 | 3.36             | 377.34           | 0.0856            | 67246       |
| 2013 | 3.55             | 395.52           | 0.0815            | 67835       |
| 2014 | 3.81             | 436.64           | 0.0800            | 62344       |
| 2015 | 3.96             | 469.68           | 0.0787            | 65499       |

Mean star rating, number of enrollments, market share for plans by year

## Problem 2

Table 2: Summary Statistics for Plans Without Star Ratings

| Year | Mean Enrollments | Mean Market Share | Total Plans |
|------|------------------|-------------------|-------------|
| 2010 | 107.34           | 0.0673            | 48643       |
| 2011 | 167.90           | 0.0900            | 12754       |
| 2012 | 172.73           | 0.0567            | 9077        |
| 2013 | 213.50           | 0.0600            | 3661        |
| 2014 | 257.60           | 0.0557            | 3708        |
| 2015 | 314.21           | 0.0405            | 4680        |

Mean for plans w/o star rating by year

### Problem 3

We can observe a spike in 2.5 star ratings in 2010 and a relatively more bell shaped distribution of star ratings for 2012 and 2015.

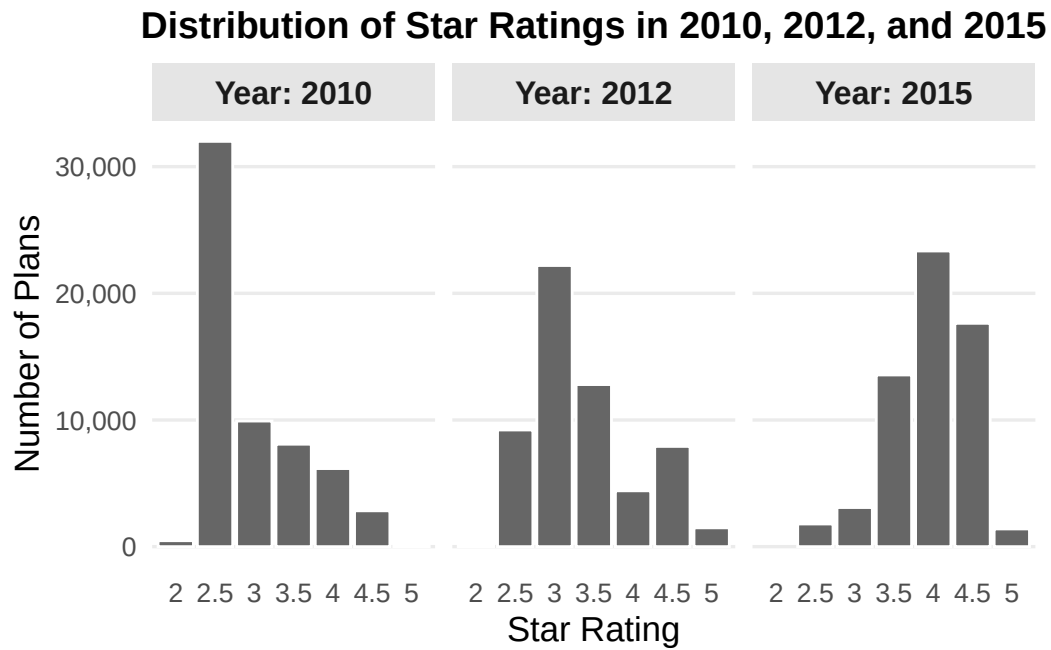


Figure 1: Distribution of Star Ratings (2010, 2012, 2015)

## Problem 4

Table 3: OLS Estimates of Star Rating Effects on Market Share

|                        | 2010    | 2011    | 2012    | 2013   | 2014   | 2015   |
|------------------------|---------|---------|---------|--------|--------|--------|
| Intercept ( 2.5 stars) | 0.0634  | 0.0702  | 0.0789  | 0.0491 | 0.0640 | 0.0606 |
| 3 Stars                | 0.0126  | 0.0350  | 0.0307  | 0.0621 | 0.0276 | 0.0271 |
| 3.5 Stars              | 0.0005  | 0.0191  | 0.0129  | 0.0385 | 0.0274 | 0.0388 |
| 4 Stars                | -0.0036 | 0.0028  | -0.0246 | 0.0172 | 0.0074 | 0.0115 |
| 4.5+ Stars             | -0.0076 | -0.0095 | -0.0134 | 0.0222 | 0.0071 | 0.0233 |

OLS Results

## Problem 5

Table 4: Plans Rounded Up to Each Star Rating (2010)

| Star Rating       | Number of Plans |
|-------------------|-----------------|
| Rounded up to 3   | 1672            |
| Rounded up to 3.5 | 1762            |
| Rounded up to 4   | 1594            |
| Rounded up to 4.5 | 193             |
| Rounded up to 5   | 29              |

Number of Plans Rounded Up

## Problem 6

Table 5: RD Estimates by Star Ratings

| Statistic     | 3 Stars  | 3.5 Stars |
|---------------|----------|-----------|
| Intercept     | 0.016    | 0.102     |
| Rounded       | 0.055    | -0.040    |
| Running Score | -0.740   | 0.417     |
| Num.Obs.      | 5309.000 | 1882.000  |
| R2            | 0.059    | 0.015     |

Estimate of Effects at 3, 3.5 Stars

## Problem 7

3 stars estimates are stable across all bandwidths, 3.5 estimates are more sensitive to bandwidth choice. Both are relatively robust as the sign of both effects do not change.

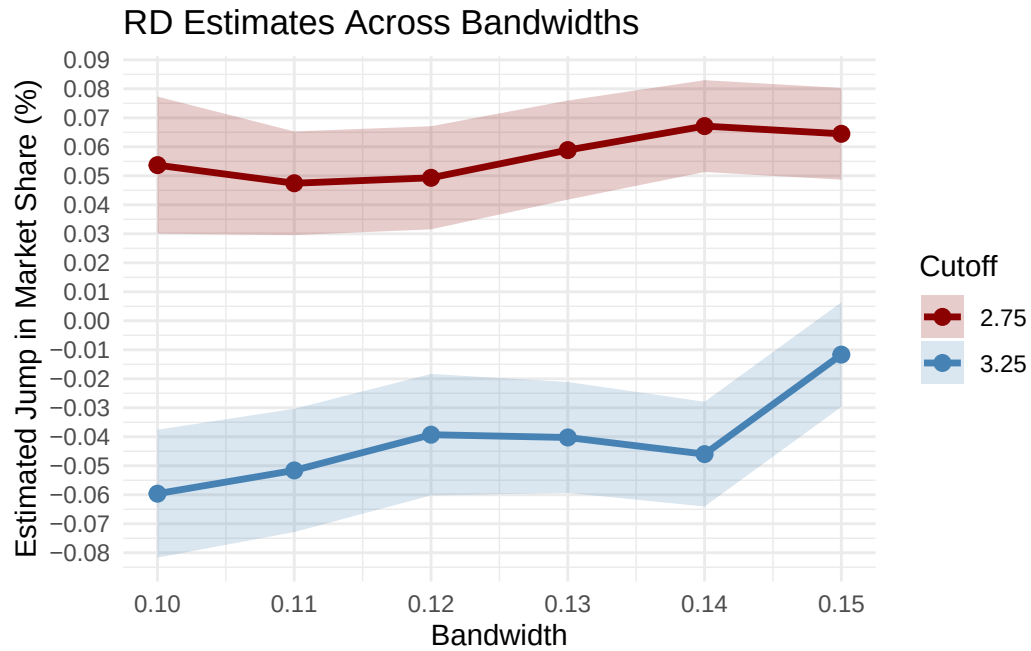


Figure 2: Different Bandwidths

## Problem 8

At the 2.75 cutoff, the density of the plans spike below the cutoff around 2.65 or -0.10 below the cutoff, showing that many contracts cluster below the threshold. However, when you look at the slope, it is downward, and market share decreases as the plan star rating is closer to the threshold. This suggests that contracts may be manipulating their scores to land below or just above the threshold. The 3 vs 3.5 star cutoff looks more balanced on both sides, with both sides having relatively flat or slightly positive slopes.

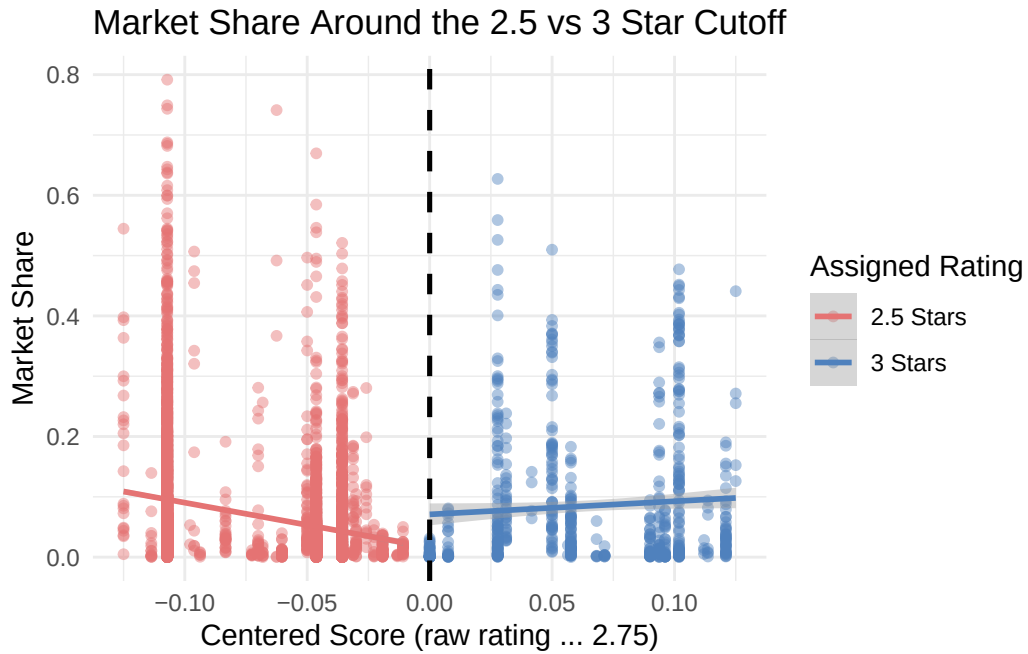


Figure 3: Distribution of Running Variable Near 2.75 Threshold



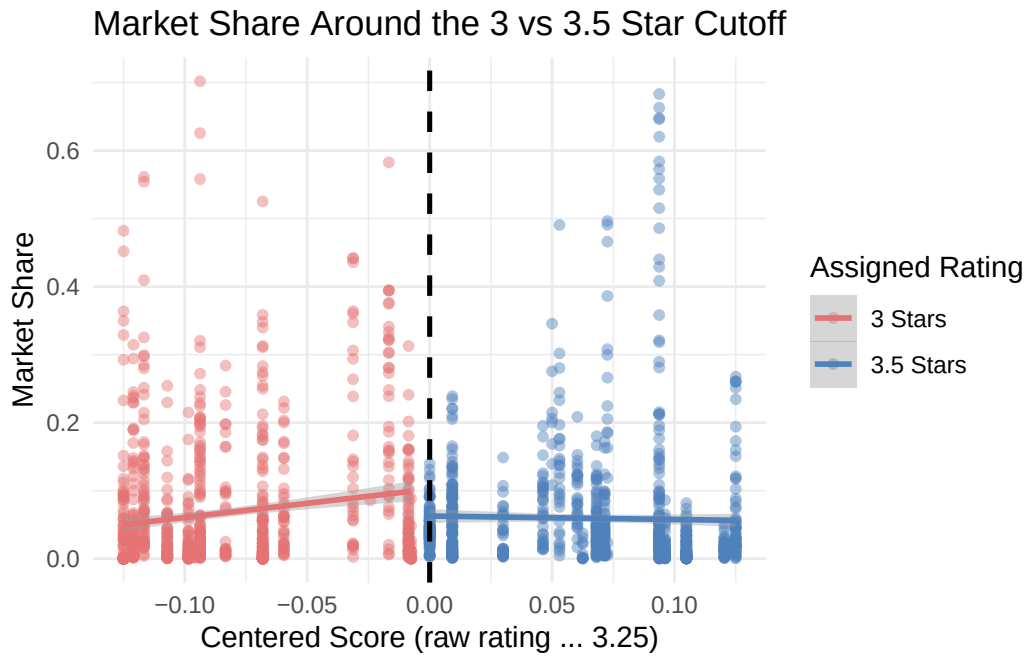


Figure 4: Distribution of Running Rariable Near 3.25 Threshold

## Problem 9

For the plans around the 2.75 cutoff, the HMO share jumps dramatically from 2.9% below to 37.5% above the cutoff, and Part D participation rises from 55.8% to 80.9%. This stark difference suggests that the plans above and below are not comparable as they do not look similar. For RD variability, we would want the plans to look similar for an ideal comparison. At the 3.25 cutoff, the HMO share percentage is quite different but the Part D participation is nearly identical, showing that the plans above and below are similar to one another.

Table 6: Plan Characteristics Around the 2.75 Cutoff (2.5-3 Stars)

| Group        | Share HMO | Share PartD | N    |
|--------------|-----------|-------------|------|
| Above cutoff | 0.3752914 | 0.8088578   | 858  |
| Below cutoff | 0.0290538 | 0.5577150   | 5094 |

### Plan Characteristics Around Cutoff

Table 7: Plan Characteristics Around the 3.25 Cutoff (3-3.5 Stars)

| Group        | Share HMO | Share PartD | N    |
|--------------|-----------|-------------|------|
| Above cutoff | 0.3798604 | 0.8484546   | 1003 |
| Below cutoff | 0.6091065 | 0.8445017   | 1164 |

### Plan Characteristics Around Cutoff

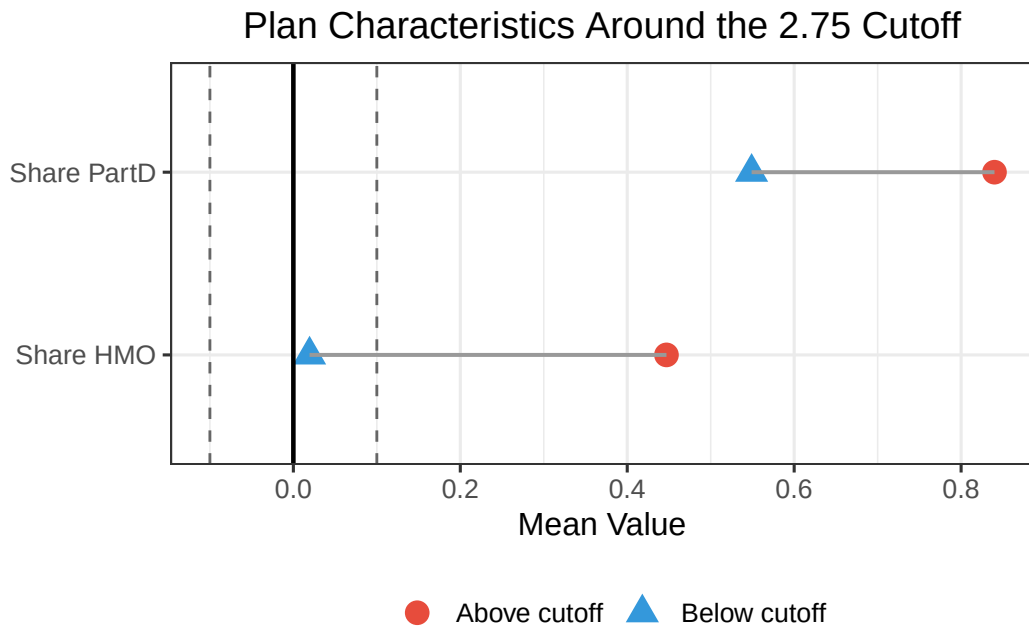


Figure 5: Love Plot for Plan Characteristics

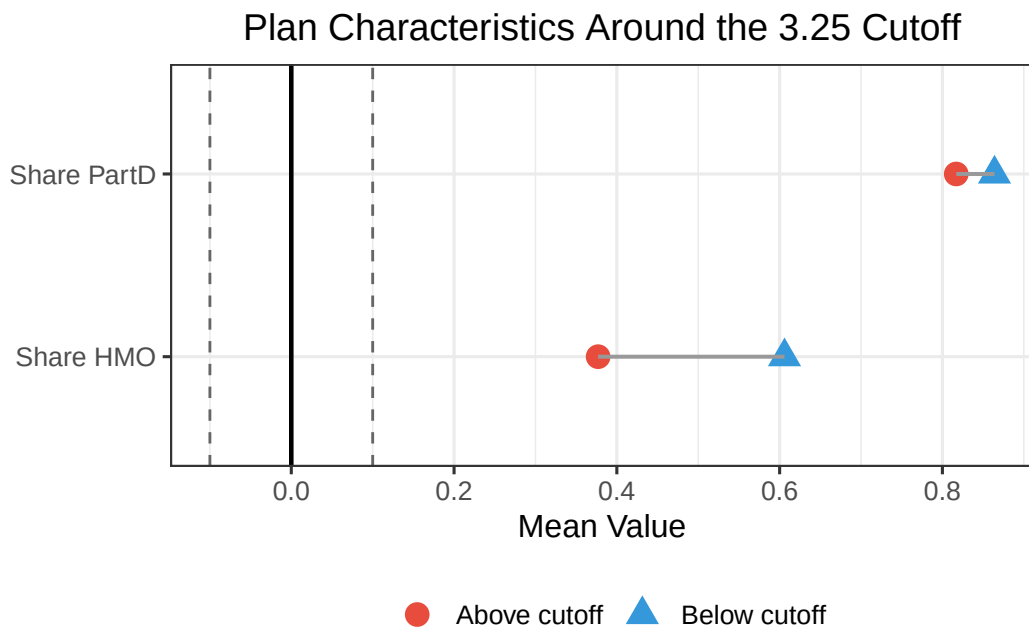


Figure 6: Love Plot for Plan Characteristics

## Problem 10

Overall, plans with higher star ratings tend to have more people enrolled in them, and this pattern holds across all years from 2010-2015. The biggest jump happens at the 2.75 cutoff because when a plan goes from 2.5 to 3 stars, it gets significantly more enrollments while the jump from 4 to 3.5 stars matter less. However, the results need to be approached with caution. First, plans are not random near the cutoff, especially around the 2.75 threshold as the slope shows a sharp increase of plans right above the threshold. Also, plans above the threshold look quite different from those below when comparing if they are HMO and/or have plan D. In conclusion, star ratings seem to matter for enrollment but we cannot quantify the direct amount of how much they matter.